8873 H1 Chemistry Preliminary Examination 2018 Answers
Section A

Answer **all** the questions in this section in the spaces provided.

1. The Earth's climate is changing. Temperatures are rising, snow and rainfall patterns are shifting, and more extreme climate events – like heavy rainstorms and record high temperatures – are already happening. Many of these observed changes are linked to the rising levels of carbon dioxide and other greenhouse gases (gases that trap heat) in our atmosphere, caused by human activities.

Gross calculation shows that while carbon dioxide accounts for 74% of the greenhouse gases, methane is the second most prevalent (14%) greenhouse gas emitted from human activities - natural gas (mainly methane) and petroleum systems activities such as incinerations, are the largest source of methane emission, accounting for 31%, while methane generated from landfills as waste decomposes, accounting for 16%.

Although methane's lifetime in the atmosphere is much shorter than carbon dioxide, but methane is more efficient at trapping radiation than carbon dioxide. Kilogram for kilogram, the comparative impact of methane is about 12 times greater than carbon dioxide.

A fossil fuel power station is a power station which burns a fossil fuel such as coal, natural gas, or petroleum to produce electricity. In an attempt to cut down carbon dioxide emissions during the production of electricity from such fuels, there has been a move in many countries to build more power stations burning natural gas rather than coal. The following table compares these two types of power station.

Type of power station	Overall efficiency of power station	Amount of by-product produced per MJ of electrical energy (1 MJ = 10^6 J)	
		SO_2	NO_2
coal	40%	0.31 g	0.64 g
natural gas	51%	0.0015 g	0.11 g

In your calculations, assume that coal consists of 95% of carbon, together with 5% of non-combustible ash.

(a) Write balanced equations for the complete combustion of

(i) methane

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$$

(ii) carbon

$$C(s) + O_2(g) \longrightarrow CO_2(g)$$

(b) The enthalpy changes of combustion of methane and carbon are :

$\Delta H_c (CH_4) = -\ 890$ kJ mol^{-1}
$\Delta H_c (C) \quad = -394$ kJ mol^{-1}

(i) Use these data to calculate how many moles of each of methane and carbon needed to be burned to produce 1 MJ of **heat** energy. [2]

Methane : $1\text{x } 10^6 / 890\ 000 = 1.12$ mol

Carbon : $1\text{x } 10^6 / 394\ 000 = 2.54$ mol

(ii) Using the efficiency figures given in the table, and your answers in **(b)(i),** calculate how many moles of each of methane and carbon needed to be burned in the power stations, to produce 1 MJ of electrical energy. [4]

Methane : 100/51 x 1.1236 = 2.20 mol

Carbon : 100/40 x 2.5381 = 6.35 mol

(c) Hence, calculate the mass of the ash that would be produced per MJ of electrical energy in a coal-fired power station. Assume that the average relative molecular mass of the ash is 220.0. [2]

Average mass of ash = 5/95 x 6.352 x 220.0 = 73.5 g

(d) Use the data in the table to suggest **two** environmental advantage of using a natural gas power station. Explain your answer. [3]

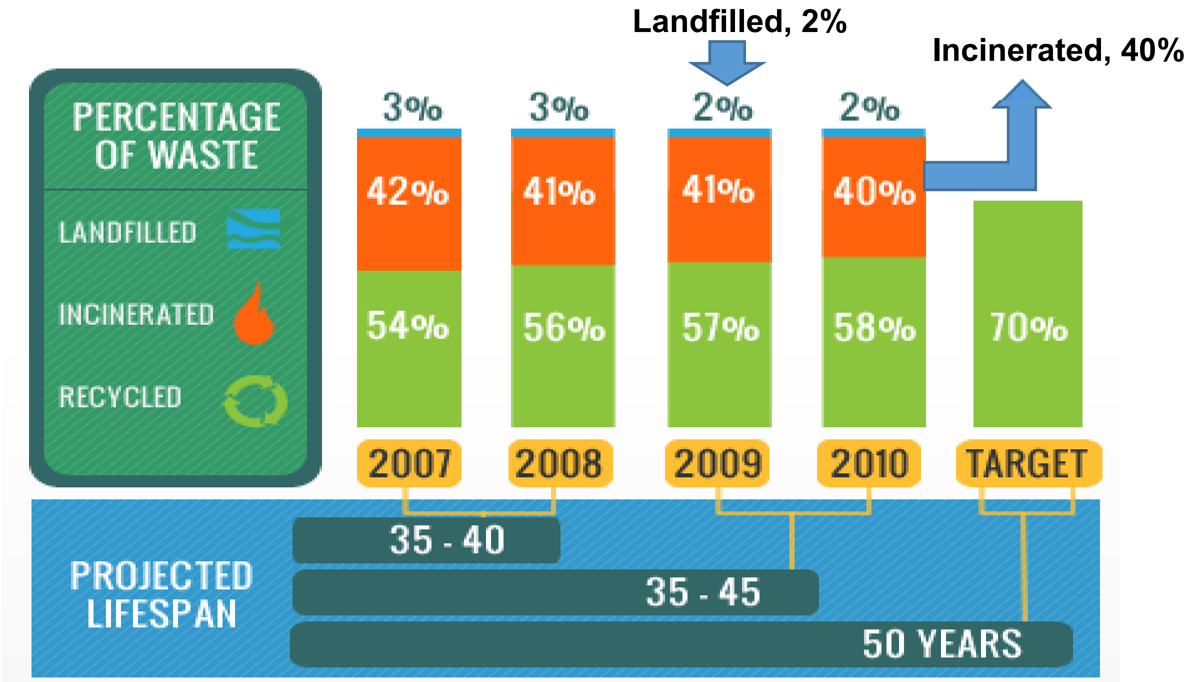
The use of natural gas in the power station produces less SO_2 and NO_2.

- **Both SO_2 and NO_2 are toxic chemicals which will cause respiratory problems**
- **Both SO_2 and NO_2 can contribute to acid rain that can cause damage to structures and harm aquatic live forms**

(e) Despite all the advantages of operating a gas power station, comment on any harmful threats behind this natural gas system. [3]

- **By the calculation in (b)(ii), 2.20 mol of methane can generate 1 MJ of electrical energy, compared to 6.35 mol of carbon**
- **As 1 mol carbon ≡ 1 mol CO_2, the use of methane is 6.35/2.20 = 2.88 times safer in terms of greenhouse gas emitted to the atmosphere**
- **However, the data given states that methane is 12 times more potent in trapping heat,**
  Hence, combining the advantage (2.88 times better in greenhouse gas emission) and disadvantage (methane is 12 times better in trapping heat), Methane is still about 4 times more potent in causing harm to the environment.

(f) Locally, the National Environment Agency, NEA has released the following data on its refuse disposal :



Use relevant data from the passage, suggest one advantage and one disadvantage for NEA to increase the use of incineration for refuse disposal over landfill. [2]

One disadvantage : **Landfill generates 16% , while incineration generates 31% of methane emission.**

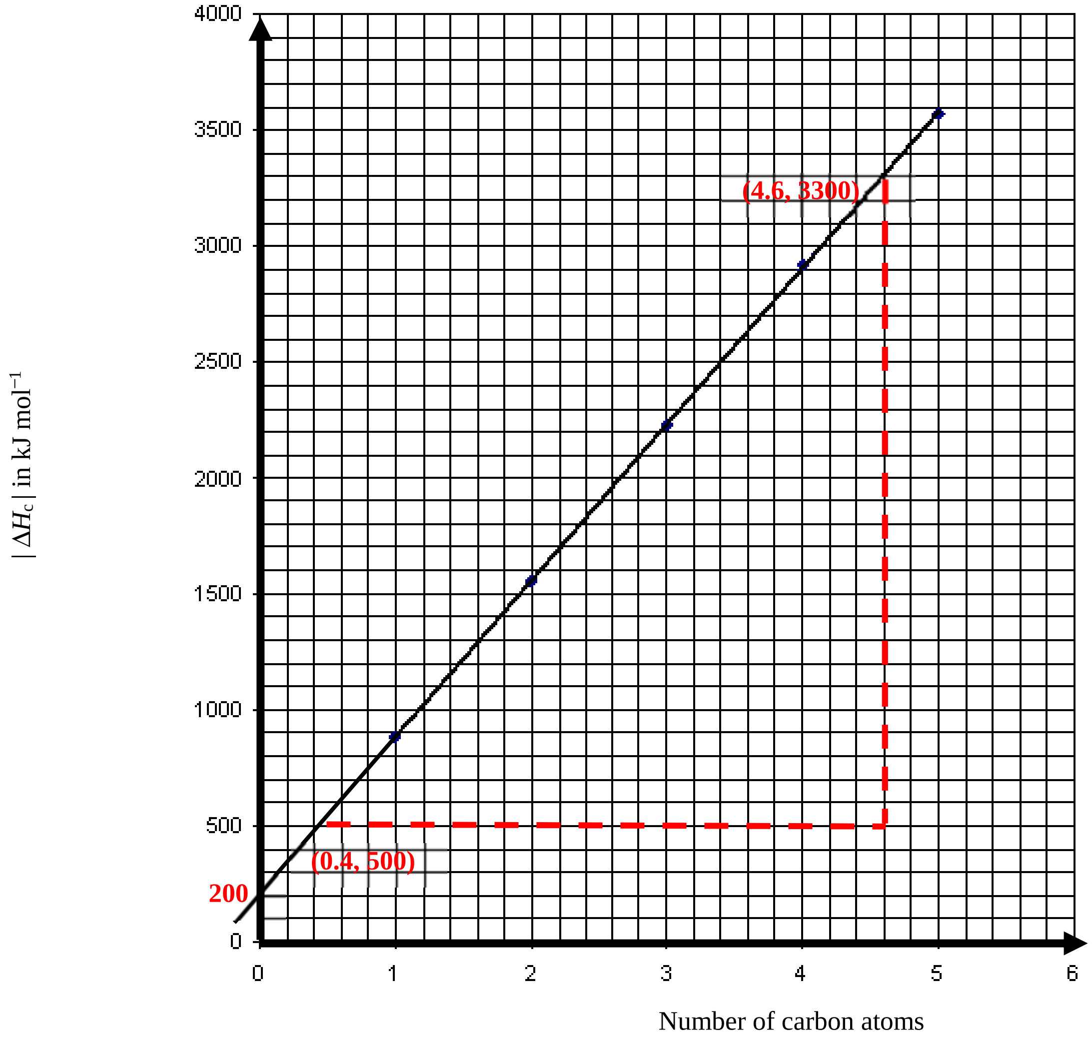
One advantage : **Incineration is favoured as Singapore is land-scarce.**

(g) Methane belongs to a family of organic compounds call Alkane.
The standard enthalpy changes of combustion of five straight chain alkanes are given in the table below.

alkane	CH_4	C_2H_6	C_3H_8	C_4H_{10}	C_5H_{12}
$\Delta H_c^{\ominus}$/ kJ mol^{-1}	–890	–1560	–2230	–2920	–3570

On the grid below, plot a graph to show how the standard enthalpy changes of combustion vary for these five alkanes. [3]

- **Either ΔH_c or $|\Delta H_c|$ can be plotted against number of carbon atoms in alkane, with the best fit line drawn through the data points**
- **Correctly labelled axes**
- **Straight line graph obtained**



Use the graph you have plotted above, determine

(i) the **average** standard enthalpy change of combustion of a "CH_2" group [2]

(ii) the standard enthalpy change of combustion of hydrogen gas [1]

(iii) the standard enthalpy change of combustion of hexane, C_6H_{14}. [2]

(g)

(i) Gradient of graph in (a) = $\dfrac{3300-500}{4.6-0.4}$ = 667 kJ mol^{-1}

So, the average enthalpy change of CH_2 group = –667 kJ mol^{-1}

(ii) Vertical intercept in (a) = 200 kJ mol^{-1}
So, $\Delta H_c^{\ominus}$ (H_2) = –200 kJ mol^{-1}

(iii) The equation of the graph is y = 667x + 200
When x = 6, [1m], y = 4200
$\Delta H_c^{\ominus}$ (hexane) = –4200 kJ mol^{-1}

(h) Alcohols are also commonly used as fuels.

The diagram below shows an energy cycle involving propan-1-ol, $CH_3CH_2CH_2OH$.

$$CH_3CH_2CH_2OH(l) + \frac{9}{2}O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(l)$$

$$\boxed{3C(s) + 4H_2(g) + 5O_2(g)}$$

Use the following data to calculate the standard enthalpy change of combustion of propan-1-ol. [2]

$\Delta H_f^\ominus(CO_2) = -393$ kJ mol^{-1}

$\Delta H_f^\ominus(H_2O) = -286$ kJ mol^{-1}

$\Delta H_f^\ominus(CH_3CH_2CH_2OH) = -303$ kJ mol^{-1}

ΔH_c + (–303) = 3(–393) + 4(–286) = –2323
ΔH_c propan-1-ol
= –2020 kJ mol^{-1}

[Total : 28]

2. The mass spectrometer made it possible to determine the masses of atoms very accurately, and it became clear that not all atoms of the same element had the same mass.

The accurate masses gave rise to the idea of the isotopes:

"Isotopes have the same atomic numbers, but different mass numbers."

(a) (i) Write the symbols showing the nucleon numbers, proton numbers and charges of the following three particles, **K, L** and **M**.

You may refer to the example given for the particle, **A** [3]

Particle	Protons	Neutrons	Electrons	Symbol
K	6	8	6	
L	7	7	10	
M	8	7	7	
A	**5**	**8**	**5**	$^{13}_{5}$**A**

(a) (i)

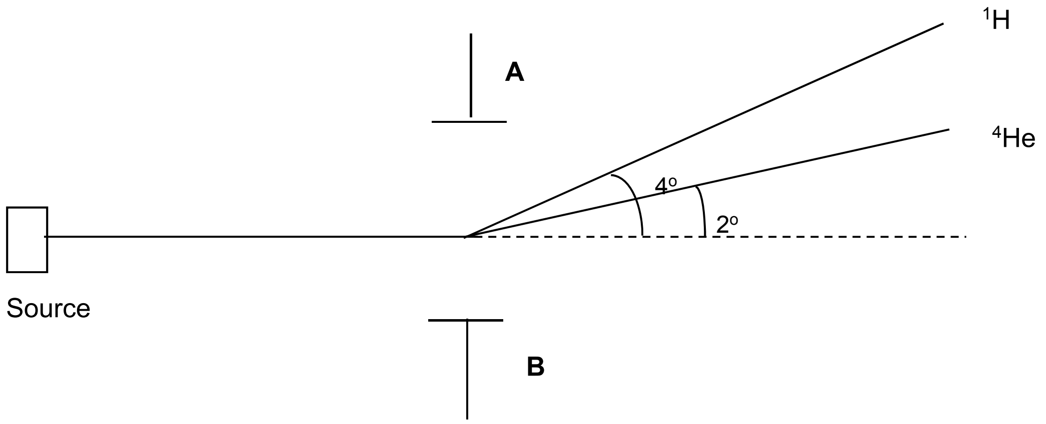
- **K:** $^{14}_{6}K$
- **L:** $^{14}_{7}L^{3-}$
- **M:** $^{15}_{8}M^{+}$

(ii) Using the data from the *Data Booklet*, write the symbols of the usual isotopes of the elements concerned. [3]

(ii)

- **Particle K:** $^{12}_{6}C$
- **Particle L:** $^{14}_{7}N$
- **Particle M:** $^{16}_{8}O$

(b) A plasma is a gaseous mixture in which the atoms have been completely stripped of their electrons, leaving bare nuclei. Because of possible use in controlled nuclear fusion reactions, plasma behaviour has been extensively studied. When passed between two plates carrying a certain electric charge, ^{1}H and ^{4}He nuclei are deflected as follows.



Giving reason for your answer, suggest the polarity (+ or -) of plate **A**. [2]

- **Plate A is negatively charged.**
- **As nuclei which are positively charged gets attracted to it.**

(c) In an analytical chemistry laboratory, the plasma torch is often used to ionise a variety of samples during their analyses.

In one series of analyses, samples of unknown elements **P** to **Y** were passed through the plasma torch and their first ionisation energies were recorded and plotted as shown.

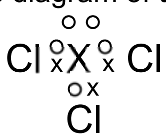
P to **Y** are consecutive elements with atomic numbers below 20.

(i) Identify the group to which element **X** belongs to, and state its electronic configuration [2]

Group : 15

Electronic configuration : $1s^2 2s^2 2p^6 3s^2 3p^3$

(ii) Give the formula of the chloride of **X**, able to function as a Lewis base and draw the dot-and-cross diagram of this chloride. [2]

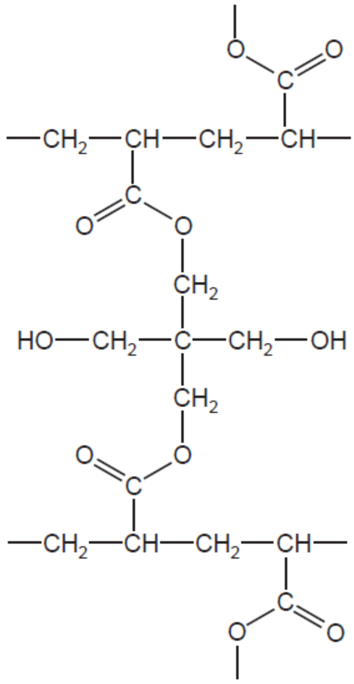


XCl_3

[Total : 12]

3. In recent years there has been considerable interest in a range of polymers known as 'hydrogels'. These polymers are hydrophilic and can absorb large quantities of water.

(a) The diagram shows part of the structure of a hydrogel.



The hydrogel is formed from the chains of one polymer which are cross-linked using another molecule.

(i) Draw the structure of a possible monomer used in the polymer chains. [1]

CH_2=CH-CO_2H <u>OR</u> CH_2=CHCO_2R <u>OR</u> CH_2=CH-COCl

(ii) State the type of polymerisation used to form these chains. [1]

Addition (polymerisation)

(iii) Draw the structure of the molecule used to cross-link the polymer chains. [1]

$C(CH_2OH)_4$

(b) Once a hydrogel has absorbed water, it can be dried and re-used many times.

Explain why this is possible, making reference to the structure shown on the previous page. [2]

Water is bonded to the polymer by <u>hydrogen bonding</u> when the hydrogel absorbs water.

<u>Hydrogen bonds are weak / easily broken</u> when the hydrogel is dried.

(c) Not every available side-chain in the polymer is cross-linked, and the amount of cross-linking affects the properties of the hydrogel.

(i) The amount of cross-linking has little effect on the ability of the gel to absorb water. Suggest why this is the case. [1]

cross-linking causes no reduction in the number of –OH groups
OR
cross-linking molecules have –OH groups

(ii) Suggest **one** property of the hydrogel that will change if more cross-linking takes place. Explain how the increased cross-linking brings about this change. [2]

property becomes harder / more rigid / less flexible / stronger / higher melting point
because the chains are more strongly / tightly held

[Total: 8]

4. In search of the perfect hair dye, materials scientists conducted experiments whereby they coated hair with a graphene solution that included water, vitamin C and a polymer to improve adhesion. They reported that the new method stayed on after 30 washings, the number necessary for a hair dye to be considered permanent.

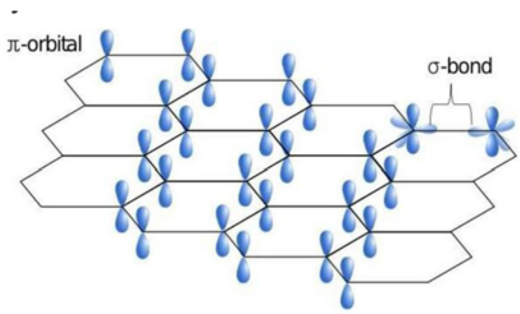
(a) State the difference between a nanomaterial and nanoparticle in terms of size. [1]

A nanomaterial is a material with at least one dimension from 1-100nm on the nanoscale while a nanoparticle is a cluster of atoms or molecules, with all three dimensions, in the range of 1–100 nm.

(b) Graphene is a nanomaterial which is strong, thin and a good conductor of electricity.

(i) Describe the structure of graphene. Explain how this structure gives graphene the properties of high strength and high electrical conductivity.

You may find it useful to include a labelled diagram in your answer. [3]



A <u>single layer of carbon atoms</u> arranged in <u>hexagons</u> whereby <u>each carbon atom bonds to 3 other carbon atoms</u> by a network of strong covalent bonds. In hexagonal lattice, each carbon has one partially filled p orbital in which the p electron is delocalised.

Description or clearly-labelled diagram

Immensely <u>strong carbon–carbon covalent bonds</u> produce an exceptionally high strength-to-weight ratio.

<u>Presence of one p electron per carbon atom delocalised</u> allows high electrical conductivity.

(ii) Suggest why a solution of graphene makes a good coating material for hair. [2]

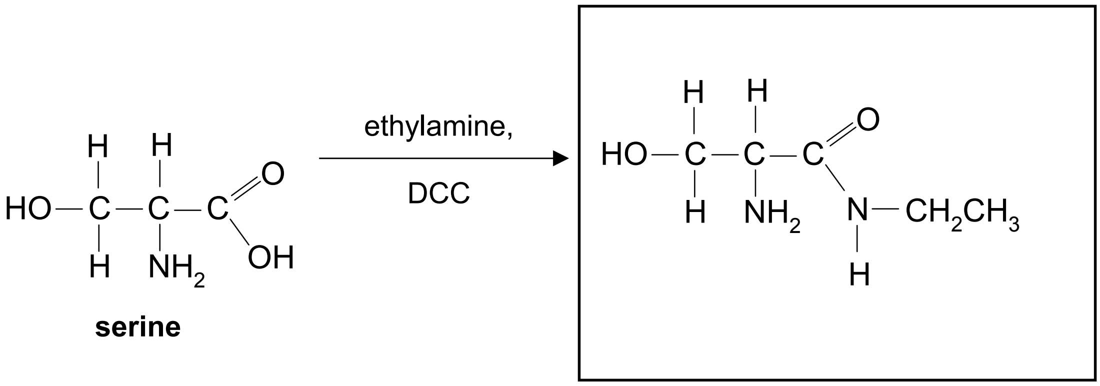
Its <u>thin, flexible sheets can adapt to uneven surfaces</u>. Also, the graphene flakes are <u>too large to absorb through the skin</u>, unlike other ingredients currently used in hair dyes.

[Total: 6]

5. Amino acids play central roles both as building blocks of protein and as intermediates in metabolism.

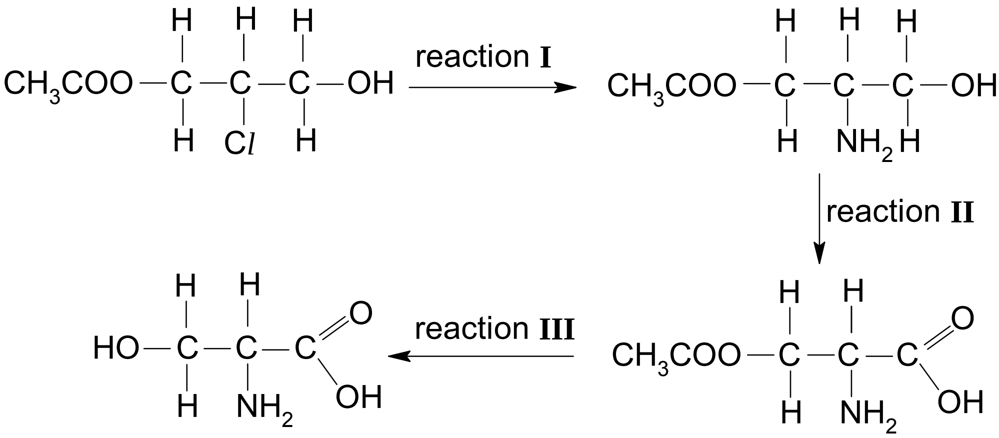
(a) Serine is a naturally-occurring amino acid.

(i) In the box below, draw the structure of the organic compound formed when serine reacts with ethylamine in the presence of dicyclohexylcarbodiimide, DCC. [1]



serine

One possible method of synthesising serine is shown below.

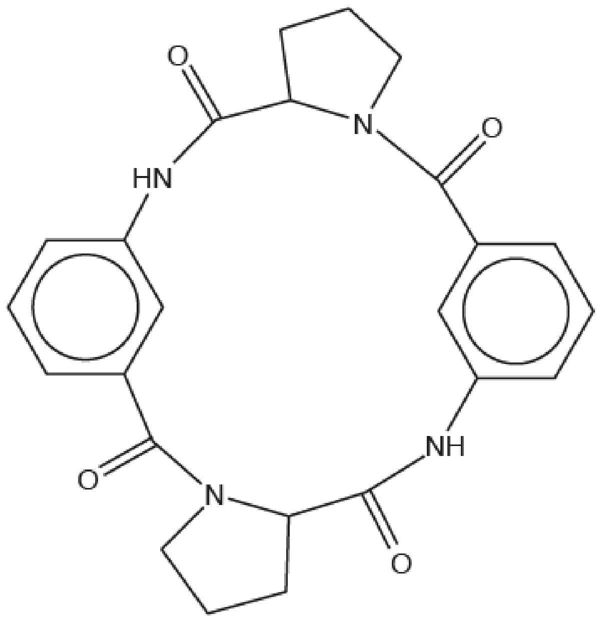


(ii) State the types of reaction for reactions **I** and **II**. [2]

reaction **I**: **substitution**

reaction **II**: **oxidation**

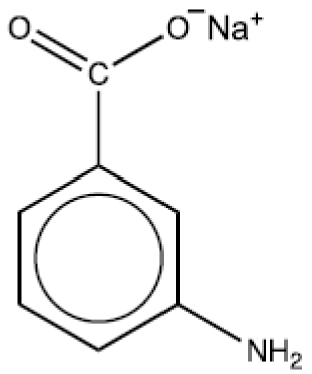
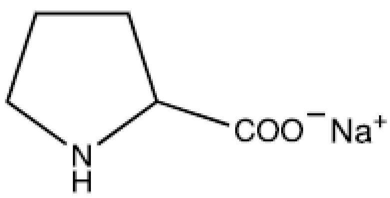
(b) A cyclic tetrapeptide has been synthesised from 3-aminobenzoic acid and an amino acid.



cyclic tetrapeptide

The cyclic tetrapeptide was treated with hot aqueous sodium hydroxide.

Draw the structures of the **two** organic products formed from the reaction and state the type of reaction undergone. [3]

type of reaction: **hydrolysis**

[Total:6]

Section B

Answer **one** question from this section on separate answer paper.

6. **(a)** Carboxylic acids are important organic compounds which are widely used as food flavourings. Carboxylic acids are usually *weak Brønsted acids*.

(i) Explain what is meant by a *weak Brønsted acid*. [2]

weak – dissociates partially in water

Brønsted acid – proton donor

(ii) The K_a values for some organic acids are listed below.

acid	K_a / mol dm^{-3}
CH_3CO_2H	1.7×10^{-5}
$ClCH_2CO_2H$	1.3×10^{-3}
Cl_2CHCO_2H	5.0×10^{-2}

Explain the trend in K_a values in terms of the structures of these acids.

[3]

- **electron-withdrawing C*l* atom disperses negative charge on $ClCH_2CO_2^-$ and $Cl_2CHCO_2^-$ ions**
- **greater number of C*l* atoms disperses negative charge on $Cl_2CHCO_2^-$ ion to a greater extent, and stabilise $Cl_2CHCO_2^-$ ion more**
- **Cl_2CHCO_2H dissociates more than $ClCH_2CO_2H$ and produces more $H^+(aq)$**

(b) Esters such as ethyl ethanoate can be formed by reacting carboxylic acids with alcohols.

$$CH_3CO_2H(l) + C_2H_5OH(l) \rightleftharpoons CH_3CO_2C_2H_5(l) + H_2O(l)$$

This reaction is allowed to proceed until a dynamic equilibrium is established.

(i) Explain what is meant by the term *dynamic equilibrium*. [1]

- **rate of forward reaction = rate of backward reaction**
- **both forward and backward reactions are on-going**

(ii) Write the expression for the equilibrium constant, K_c, for this reaction. [1]

$$K_c = \frac{[CH_3CO_2C_2H_5][H_2O]}{[CH_3CO_2H][C_2H_5OH]}$$

(iii) For this equilibrium, the value of K_c is 4.0 at 298 K.

A mixture containing 0.5 mol ethanoic acid, 0.5 mol ethanol, 0.1 mol ethyl ethanoate and 0.1 mol water was set up and allowed to come to equilibrium at 298 K.

Calculate the amount, in moles, of each substance present at equilibrium. [3]

$$CH_3CO_2H(l) + C_2H_5OH(l) \rightleftharpoons CH_3CO_2C_2H_5(l) + H_2O(l)$$

I	**0.5**	**0.5**	**0.1**	**0.1**
C	**-x**	**-x**	**+x**	**+x**
E	**0.5 – x**	**0.5 –x**	**0.1 + x**	**0.1 + x**

- **Let change in concentration be *x***
  At equilibrium, $[CH_3CO_2H] = [C_2H_5OH] = 0.5 - x$
  and $[CH_3CO_2C_2H_5] = [H_2O] = 0.1 + x$
- **So,** $\left(\frac{0.1+x}{0.5-x}\right)^2 = 4.0$ **i.e. $x = 0.30$**
- **equilibrium $[CH_3CO_2H] = [C_2H_5OH] = 0.20$ mol dm^{-3}**

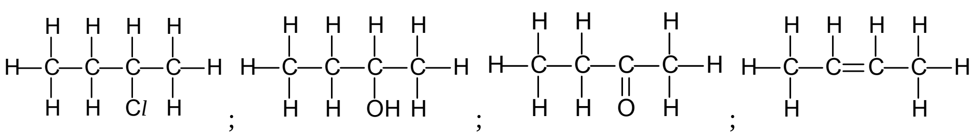
  equilibrium $[CH_3CO_2C_2H_5] = [H_2O] = 0.40$ mol dm^{-3}

(iv) State and explain what would happen to the position of the equilibrium when 0.05 mol ethyl ethanoate is withdrawn from the equilibrium mixture in **(iii)**. [2]

- **the equilibrium position shifts to the right**
- **to replenish $CH_3CO_2C_2H_5$**

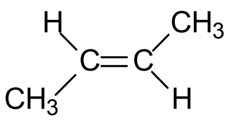
(c) **H** is a halogenoalkane with molecular formula C_4H_9Cl. When reacted with hot ethanolic NaOH, the product **J** is found to display cis-trans isomerism. When refluxed with aqueous NaOH, compound **K** is formed.
K undergoes oxidation to form product **L**. **L** does not react with Tollens' reagent.

(i) Using the information given above, suggest the structures of compounds **H,J,K** and **L.** [4]

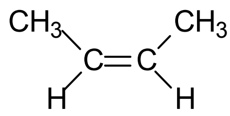


compound H ; **compound K** ; **compound L** ; **compound J**

(ii) Draw and name the two isomers of compound **J**. [2]



trans-but-2-ene



cis-but-2-ene

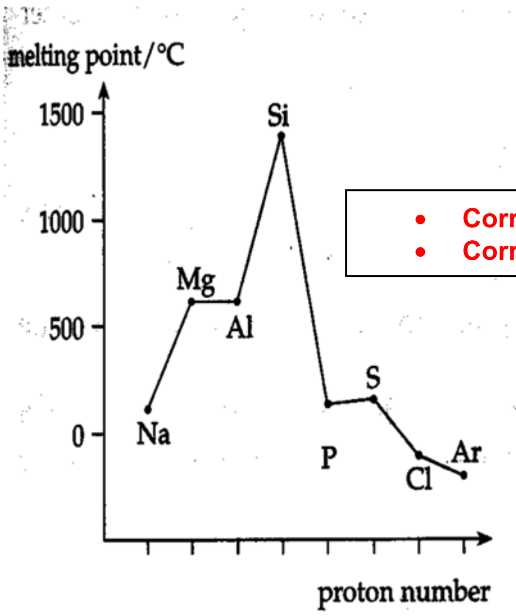
(iii) Compound **K** is found to have a higher boiling point as compared to compound **L**. With reference to structure and bonding, explain why this is so. [2]

- **Compound K has a <u>simple molecular structure</u> with <u>intermolecular hydrogen bonds</u>**
- **compound L has a simple molecular structure with <u>pd-pd attraction between molecules</u>**
- **<u>More energy</u> is needed to break the <u>stronger</u> hydrogen bonds between molecules of K compared to the <u>weaker</u> pd-pd forces of attraction between molecules of L . Hence, K has a higher boiling point than L**

[Total : 20]

7. Phosphorus, P, is an element found in the third period of the Periodic Table. The element was discovered by Hennig Brand when he experimented with urine in 1669.

(a) **(i)** Sketch a graph showing the variation of melting point of the elements across the third period of the Periodic Table. [2]



- **Correctly labelled axes**
- **Correct shape of sketch**

(ii) Explain qualitatively the shape of the graph you have drawn in **(a)(i)**. [2]

- **Na, Mg, A*l* : All have high mp. Giant metallic structures w/ strong electrostatic forces of attraction between metal cations and delocalized ē.**
- **Si : Very high mp. Giant molecular structure w/ numerous strong covalent bonds between atoms.**
- **P, S, C*l* : Low mp. Simple molecular structures w/ weak id-id forces between the molecules for P_4, S_8, Cl_2 , and Simple atomic structure w/ weak id-id forces between the atoms for Ar.**

Strength of id-id forces of attraction α number of electrons.

Order of melting point: $S_8 > P_4 > Cl_2$

Magnesium and aluminium are also elements from the third period of the Periodic Table. Both elements react with excess oxygen gas to form their respective oxides, MgO and Al_2O_3.

At room temperature, both MgO and Al_2O_3 are white crystalline solids. These compounds have useful applications in our daily lives – MgO is used as a drying agent to preserve books and Al_2O_3 is used as an abrasive in sandpapers.

(b) A sample of white powder is known to be either MgO or Al_2O_3. In order to determine its identity, **one** of the following reagents can be used.

- dilute hydrochloric acid
- aqueous sodium hydroxide
- aqueous hydrogen peroxide

(i) State the reagent you would use to determine the identity of the white powder. [1]

NaOH(aq)

(ii) Explain your choice in **(b)(i)**, and write balanced equations for any chemical reactions that may occur. [3]

- **Al_2O_3 is able to dissolve in NaOH(aq)**
- **but MgO will not**
- **$Al_2O_3 + 2OH^- + 3H_2O \rightarrow 2Al(OH)_4^-$**

Arsenic, As, is in the same Group of the Periodic Table as phosphorus. The element burns in excess oxygen gas to form arsenic(III) oxide, As_2O_3. An acidified solution of vanadium(V) ions is able to gradually oxidise arsenic(III) oxide to arsenic(V) oxide, according to the following balanced equation.

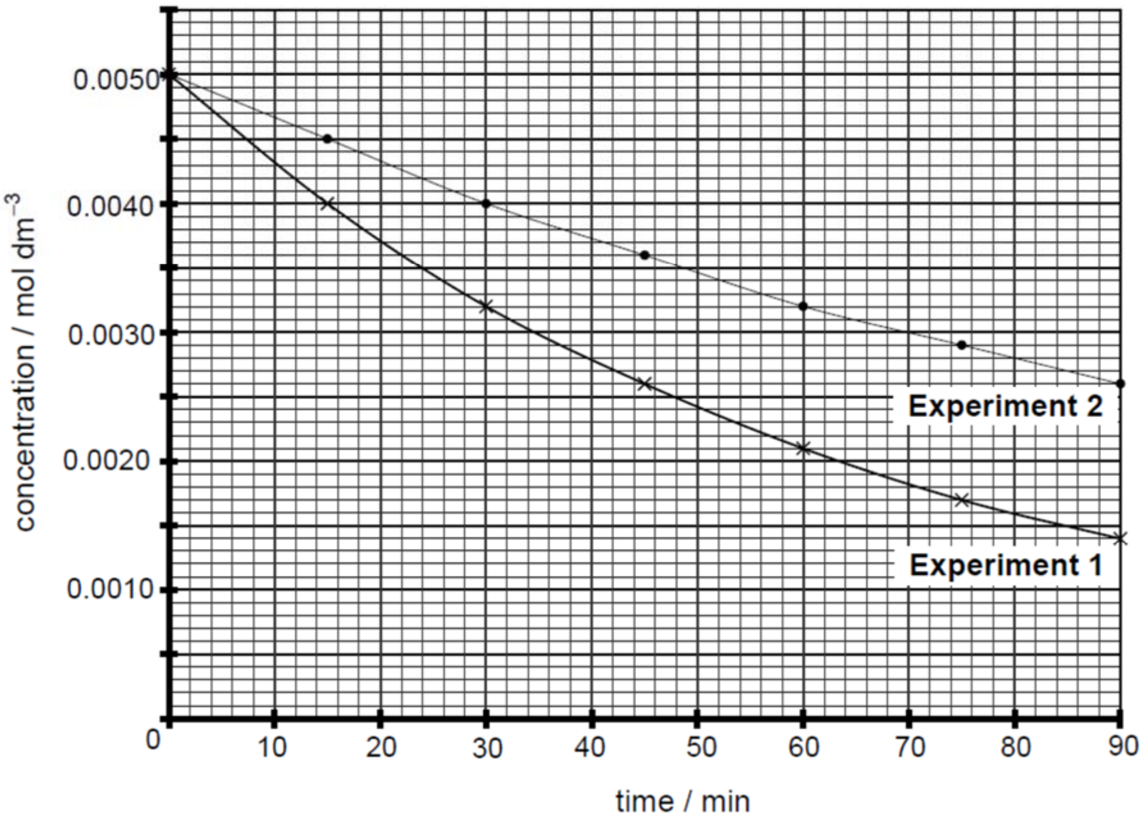
$$As_2O_3 + 4VO_2^+ + 4H^+ \rightarrow As_2O_5 + 4VO^{2+} + 2H_2O$$

(c) The rate of the reaction between acidified VO_2^+ ions and arsenic(III) oxide at 25°C was followed by measuring the concentration of remaining vanadium(V) ion after fixed time intervals. Two experiments were carried out, starting with different concentrations of arsenic(III) oxide.

Experiment 1: $[As_2O_3] = 0.10$ mol dm^{-3}

Experiment 2: $[As_2O_3] = 0.05$ mol dm^{-3}

The following graphs show the results of the two experiments.



(i) Define *order of reaction*. [1]

The order of reaction with respect to a particular reactant is the power to which the concentration of the reactant is raised to in the rate equation.

(ii) Use the graphs to determine the order of reaction with respect to arsenic (III) oxide, As_2O_3 and vanadium (V) oxide ion, VO_2^+ respectively. [4]

Let Rate = $k[VO_2^+]^m[As_2O_3]^n$

Since the reaction has a constant half-life ($t_{1/2}$ = 48 min) ,
the reaction is first order with respect to VO_2^+. i.e. $m = 1$

Rate = $k[VO_2^+]^1[As_2O_3]^n$

Initial rate of experiment 1 = $\left|\dfrac{0.0050-0.0022}{0-40}\right|$ = 7.0×10^{-5} mol dm^{-3} min^{-1}

Initial rate of experiment 2 = $\left|\dfrac{0.0050-0.0036}{0-40}\right|$ = 3.5×10^{-5} mol dm^{-3} min^{-1}

Experiment	Initial $[VO_2^+]$ / mol dm^{-3}	Initial $[As_2O_3]$ / mol dm^{-3}	Initial rate / mol dm^{-3} min^{-1}
1	0.0050	0.10	7.0×10^{-5}
2	0.0050	0.05	3.5×10^{-5}

[1m]

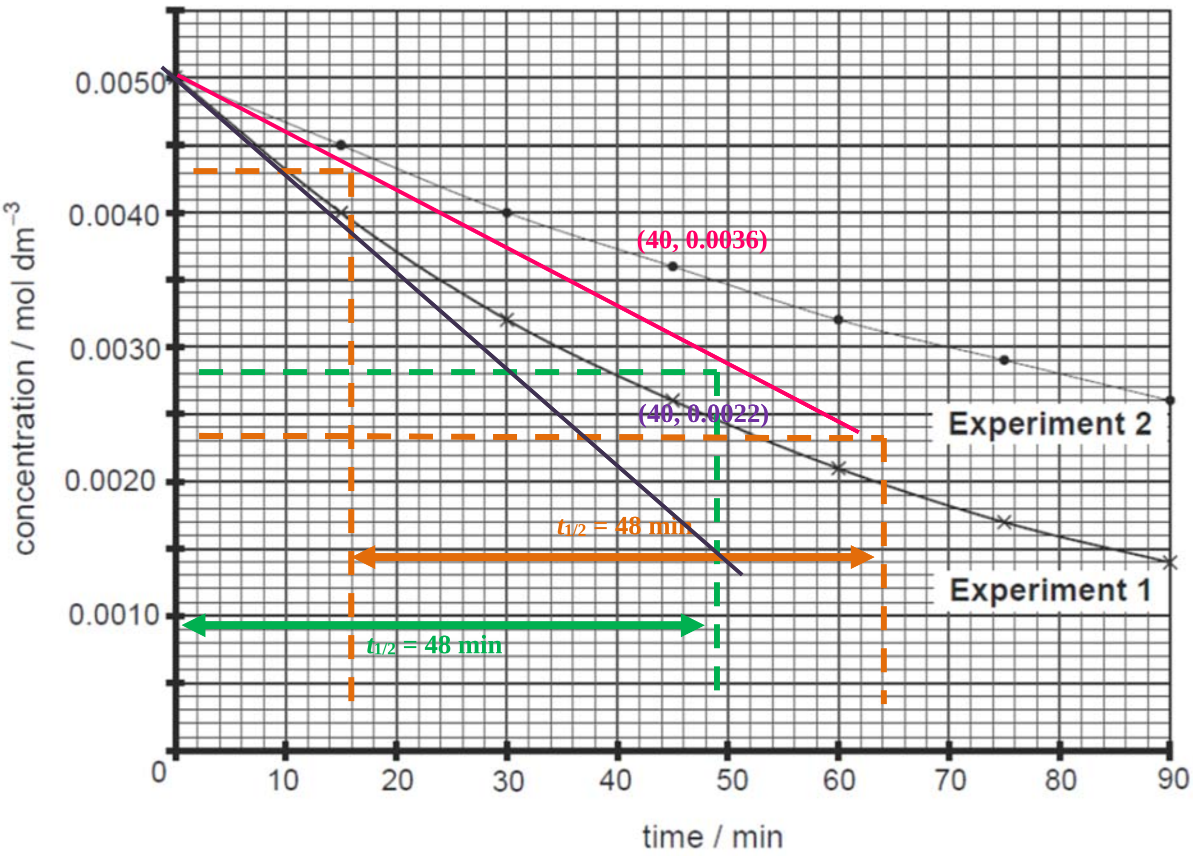
Comparing experiments 1 and 2: $\dfrac{7.0\times\cancel{10^{-5}}}{3.5\times\cancel{10^{-5}}} = \dfrac{\cancel{k}(0.0050)(0.10)^n}{\cancel{k}(0.0050)(0.05)^n}$

$2 = \left(\dfrac{0.10}{0.05}\right)^n$

$2 = 2^n$ i.e. $n = 1$

(iii) Write the rate equation for the reaction. [1]

Rate = $k[VO_2^+][As_2O_3]$



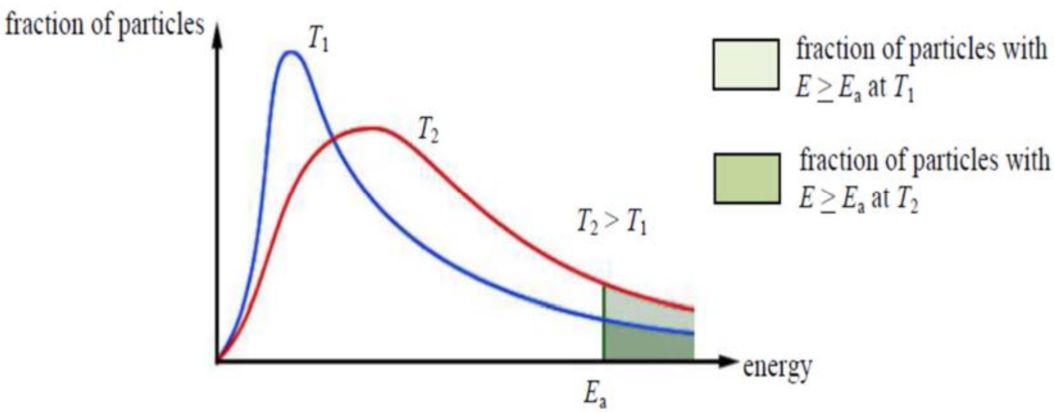
(iv) Calculate the rate constant for the reaction, stating its units. [2]

From experiment 1: $k = \dfrac{7.0\times10^{-5}}{0.0050\times0.10}$

= 0.14 mol^{-1} dm^3 min^{-1}

(v) A third experiment involving the reaction between As_2O_3 and VO_2^+ is carried out at 40°C.

With the aid of a suitable diagram, state and explain how the rate of reaction at 40°C would compare with that at 25°C. [4]



- **When temperature increases, the reactant particles have higher kinetic energies. Hence, there will be a higher frequency of effective collisions.**

- **When temperature increases, a greater fraction of the particles will have energy greater or equal to the activation energy required for reaction.**
  Rate constant will increase at a higher temperature
- **Rate of reaction at 40^0C is greater than that at 25^0C**